Tectonophysics, 60 (1979) 169—188

© Elsevier Scientific Publishing Company, Amsterdam — Printed in The Netherlands

PALEOMAGNETIC EVIDENCE FOR NON-APENNINIC ORIGIN OF THE SILA NAPPES (CALABRIA)

M. MANZONI

Laboratorio per la Geologia Marina, National Research Council of Italy, Bologna (Italy)

(Received May 21, 1978; revised version accepted November 6, 1978)

ABSTRACT

Manzoni, M., 1979. Paleomagnetic evidence for non-Apenninic origin of the Sila nappes (Calabria). Tectonophysics, 60: 169—188

The magnetization of Lower Permian rocks from Sila has a mean direction $D = 56.5^\circ$, $I = +20.4^\circ$ with $\alpha_{95} = 9.1^\circ$ after correction for Upper Neogene tilting. A further correction for the attitude of the nappes after their Middle Miocene emplacement establishes paleolatitudes consistent with those from the Lower Permian Tethys. The remarkable internal consistency of the data has not supported the distinction of units with opposite vergences within the Sila crystalline nappes. The declination indicates that the Sila massif has rotated counter-clockwise by about 90° relative to the Apennines, Sardinia and the Southern Alps and therefore the well-known Apenninic rotation alone does not account for the total change of direction in tectonic transport. Accordingly, the structural trends of tectonic phases older than the emplacement time of the Calabrian nappes should no longer be referred to present-day geographic coordinates. The post-Late Cretaceous motion relative to the north Calabrian Apennines enhances the geotectonic role of the northern boundary of the Calabrian—Peloritan arc, since its sinistral-shear character permits both tectonic transport from the west and counter-clockwise motion during tectonic transport.

INTRODUCTION

The Calabrian—Peloritan arc links the southern Apennines with Sicily. Its structural association to the Apenninic chain has been questioned or denied by many and its origin is one of the most debated issues in the interpretation of the structure of the Mediterranean.

The Sangineto line marks its northern boundary in Calabria and the Taormina line marks its southern boundary in Sicily; both lines have been assumed by diverging modern interpretations to represent the Africa—Europe boundary.

THE TECTONIC SETTING

The allochthonous character of all the exposed pre-Miocene rocks piled up in complex nappes is now generally accepted. The occurrence among them

of nappes with Hercynian plutonites and metamorphic rocks (Calabride units by Ogniben, 1969) has fostered divergent speculations on the original location, the history of deformation, and the transport of the fragmented continental crust.

In recent interpretations some models consider all structures as originally belonging to an Apenninic domain or geo-syncline with most, but not all, directions of transport from the west (e.g. Ogniben, 1969).

Dubois, 1976 and other authors assume that a north-African chain was fragmented, deformed and displaced by the Europe—Africa collision. Microplate models advocate the drifting of the entire crust of Calabria from a fragmented Alpine belt east of Sardinia, in spite of the known allochthony of all exposed terreins and of the lack of information on the nature of the basement (Alvarez et al., 1974).

When units from an Alpine chain are considered as the source of the upper nappes with Hercynian rocks, such as the Sila nappes, coming to override the structures of the Apenninic chain, different phases of directions of transport are required.

According to Haccard et al. (1972): there was first an Alp—Apennine compression, then tectonic transport from the east, and subsequently Apenninic transport from the west, the southwest and the south. According to Mascle and Grandjacquet (1978), Cretaceous transport from the south or southeast and east was followed by Upper Cretaceous east—west stresses, then by four phases of tectonic activity with Apenninic vergence.

Even the basic assumptions required by different models may be mutually exclusive: the empilement of nappes in a former western Alpine chain, later transported as a whole toward the east by the Middle Miocene Apenninic phase (e.g., Ogniben, 1973) is not compatible with models where the nappes are transported from other directions into Calabria to override already emplaced nappes of the Apenninic chain. In the first case the Alpine directions of transport may remain undetermined in terms of present geography; in the second case, directions are chosen according to the requirements of the proposed models. The complexities of the interpretations stem from an extremely intricate geologic history.

Amodio Morelli et al. (1976) have made an interpretative review of models put forward to explain the origin of the Calabrian—Peloritan arc. In this reconstruction the Alpine nappes with vergence toward Europe overrode Apenninic terrain with southeasterly and easterly vergence toward the African foreland during the Miocene. The upper nappes with Hercynian terrains from the Sila massif are split into two units with opposite vergences: a lower Longobucco—Longi—Taormina nappe with African vergence and an overlying "incertae sedis" Mt. Gariglione unit (Fig. 1).

Both units are made up of plutonites (granodiorites, tonalites, porphyritic and aplitic dykes) with associated high and medium-grade metamorphites (migmatites, amphibolites, gneisses, gabbroids). The Longobucco—Longi—Taormina unit also displays a wider compositional range of dykes, low-grade

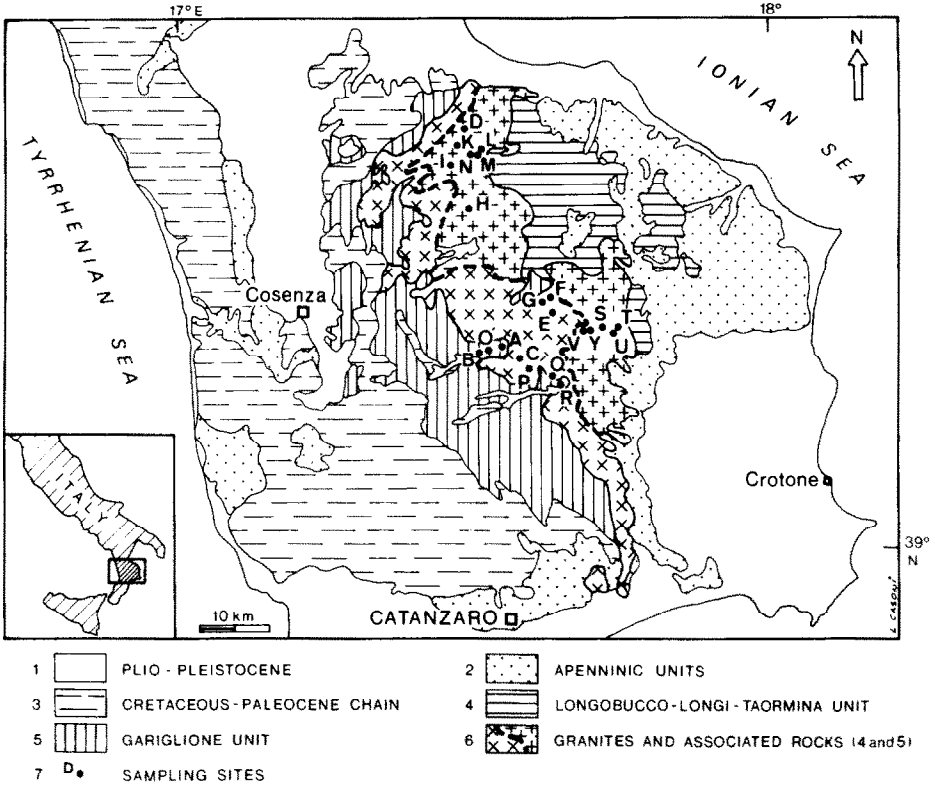


Fig. 1. Geologic sketch of the Sila sampling area (redrawn after Bonardi et al., 1976).

metamorphites and a Mesozoic sedimentary cover.

The only absolute-age determination for the Sila units to date is 260 ± 10 m.y. (Lower Permian), on biotite from a granodiorite of the Gariglione unit (Borsi and Dubois, 1968). It corresponds to the age of the Hercynian plutonism in Sila, but it should not be extrapolated to the wide range of plutonites found in the nappes. Paleomagnetic data from the upper Calabride units from Sila sought primarily to make an independent evaluation of allochthony and of the degree of tectonic disruption within the crystalline complex.

SAMPLING, TESTS AND RESULTS

Diorites, amphibolites, mafic xenoliths within the granites, and basic to acidic dykes from a 20-km long belt yielded consistent directions of stable remanent magnetization at 17 sites out of 21 after a preliminary collection of one to three oriented samples per site. In the final sampling at least five hand-samples per site were collected from 15 selected sites only. Most sam-

TABLE I

Site mean directions and precision parameters from Sila (Calabria)

Site	N.R.M.						100 Oe				
	n	D	I	α_{95}	R	Int. 10^{-5}	n	D	I	α_{95}	R
A	10	29.5	+50.5	14.4	9.261	0.8	10	42.5	+53.9	15.2	9.181
C	6	48.3	+42.5	12.2	5.837	2.9	6	55.2	+39.1	11.4	5.858
E	7	39.9	+41.3	30.2	5.782	4.8	6	60.1	+46.4	15.3	5.751
H	9	17.2	+64.0	22.1	8.639	63.0	—	—	—	—	—
N	8	25.7	+35.8	22.4	7.004	0.8	7	35.9	+22.0	6.3	6.934
O	8	59.3	+57.1	16.0	7.458	3.3	—	—	—	—	—
P	10	63.5	+47.7	23.0	8.324	16.9	8	54.6	+40.7	19.5	7.223
Q	7	268.8	+18.4	11.4	6.792	10.1	7	293.3	+46.6	33.2	5.586
R	8	53.3	+11.8	12.2	7.672	3.3	—	—	—	—	—
T	10	56.0	+29.0	15.1	9.189	2.6	9	63.7	+25.4	11.3	8.631
U	6	40.9	+46.1	22.3	5.498	0.7	6	53.9	+33.5	26.0	5.340
X	8	71.4	+38.4	5.4	7.934	9.2	—	—	—	—	—
Y	7	28.3	+72.1	41.7	5.028	14.0	6	60.9	+64.9	32.4	5.044

Overall mean directions:

n	D	I	α_{95}	R	
11:	54.2	+40.4	9.5	10.584	(excluded sites H and R)
13:	55.8	+39.6	11.7	12.100	(all sites).

ples from sites B and D did not bear stable remanence. The locations of the sites are shown in Fig. 1, redrawn after Bonardi et al. (1976).

Sampling was made (1) in those parts of the intrusive masses richer in mafic minerals (diorites and gabbroids at sites A, C, O); (2) in the amphibolic xenolyths occurring in large numbers in several areas within the gneisses and the granites (site T); and (3) from dykes, i.e. lamprophyric dykes (site H), basic fine-grained dykes (sites N, Q, U, X, Y), "syenitic" dykes (sites F, R), acidic fine-grained dykes (sites I, L, M, S), and porphyritic dykes (sites D, E, G, K).

Over 260 specimens from 118 hand-samples were measured and subjected to AF demagnetization tests in the C.N.R. paleomagnetic laboratory in Bologna.

At least one pilot specimen per site was subjected to stepwise AF demagnetization and subsequently all specimens were cleaned in four steps. The results for the demagnetizing fields yielding the tightest clusters of directions were selected for all 13 sites (Table I and Fig. 4).

On the basis of the response to AF cleaning, the remanent magnetization may be classified into three types. Nine sites carried secondary components which could be removed by AF between 50 and 400 Oe (Figs. 2 and 3a,

200 Oe					400 Oe					Site
n	*D*	*I*	α_{95}	*R*	*n*	*D*	*I*	α_{95}	*R*	
10	47.3	+54.0	15.0	9.205		—	—			A
6	59.1	+38.2	11.0	5.868	6	66.4	+42.3	13.3	5.810	C
6	59.1	+47.4	17.4	5.682		—	—	—	—	E
9	29.9	+71.1	5.4	8.911	9	65.7	+73.6	3.9	8.953	H
7	35.4	+20.9	8.9	6.871	7	33.4	+30.6	8.5	6.884	N
8	46.8	+60.9	12.2	7.674	8	32.7	+62.3	27.5	6.597	O
8	65.1	+31.5	23.7	6.905	8	62.5	+24.6	28.2	6.538	P
7	38.3	+30.1	50.5	4.476	7	35.9	+32.8	17.8	5.608	Q
—	—	—	—	—	8	65.7	—5.6	8.5	7.839	R
9	64.8	+25.2	12.8	8.533	—	—	—	—	—	T
6	46.1	+25.9	16.1	5.727	6	44.4	+49.0	16.4	5.713	U
8	82.8	+38.1	5.3	7.935	8	80.1	+37.3	5.1	7.941	X
6	64.7	+56.2	17.3	5.687	6	65.6	+52.7	5.7	5.898	Y

type 1). Two sites exhibited no apparent secondary components (Fig. 3b, type 2). Two sites exhibited remanence which was destroyed by AF higher than 150 Oe (type 3). In all pilot specimens remanent magnetization was destroyed in 800—1200 Oe AF with the onset of intense viscous components of magnetization.

A mean paleomagnetic direction $D = 55.8°$, $I = +39.6°$ was calculated for 13 sites. Excluding sites H and R, where deviating inclinations are regarded as evidence for severe local tilting (field evidence for faulting having been found near site R), the mean direction of 11 sites does not change significantly and becomes $D = 54.2°$, $I = 40.4°$ with $\alpha_{95} = 9.5°$.

Similar northeasterly declinations and low-positive inclinations were also found after cleaning at sites G, K, M, S, where too few samples were collected to calculate statistical parameters.

DATA CORRECTION AND INTERPRETATION

The tight grouping of the paleomagnetic directions argues against widespread large-scale tectonic disruption within the crystalline complex of Sila. Although the Mesozoic sedimentary cover is often severely deformed, the

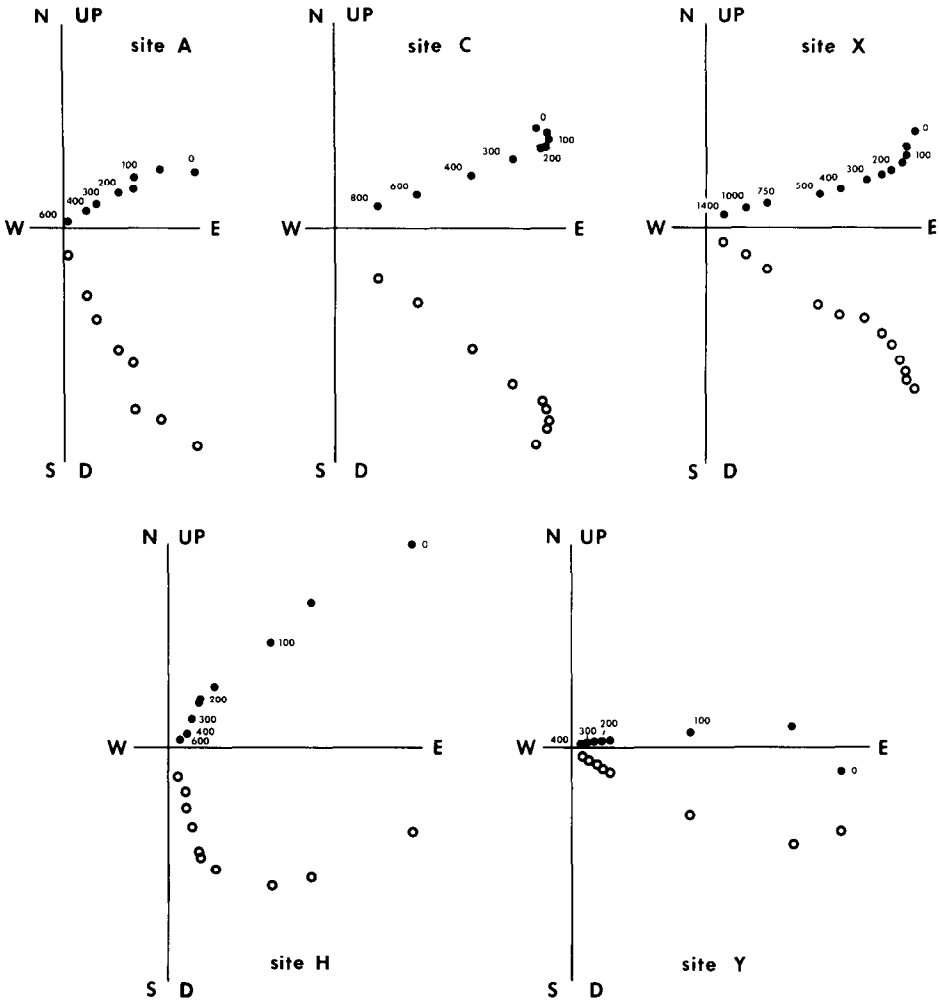


Fig. 2. Orthogonal projections of the vector paths of remanent magnetization during AC cleaning for pilot specimens from sites A, C, H, X and Y. Numbers denote Oe peakvalues of applied alternating fields.

greater strength of the crystalline slabs seems to have better resisted internal deformation.

A minor difference can be seen between data from the two tectonic units: the directions of the rocks assigned to the Mt. Gariglione unit are more tightly grouped and approach more closely the overall mean direction of magnetization, whereas those from the Longobucco—Longi—Taormina unit are more scattered, as if this unit were more severely disrupted, and they show a trend toward slightly lower inclinations. However, these variations are smaller than the radii of the cones of confidence, so that the subdivi-

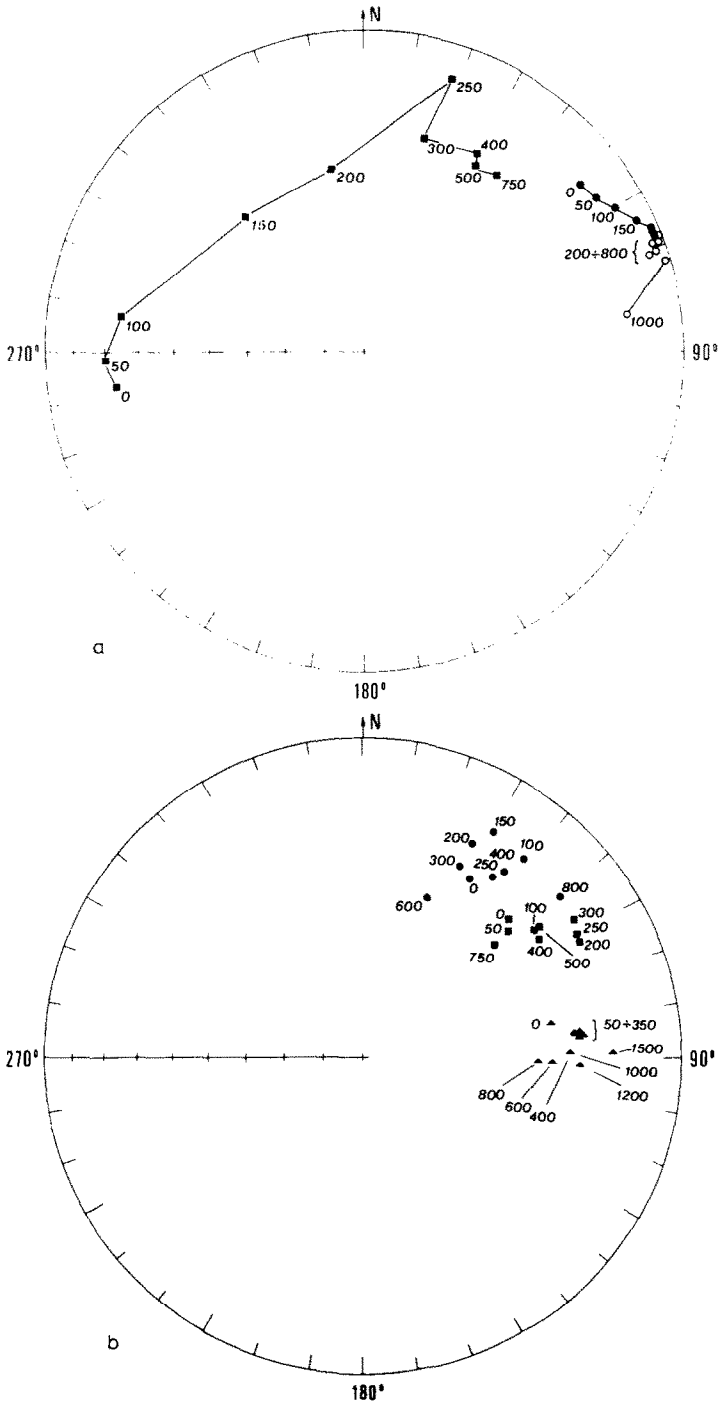


Fig. 3. Stereographic projections of the vector paths of remanent magnetization during AC cleaning for pilot specimens from sites Q (squares), R (circles) (Fig. 3a), and for pilot specimens from sites U (circles), O (squares), E (triangles) (Fig. 3b).

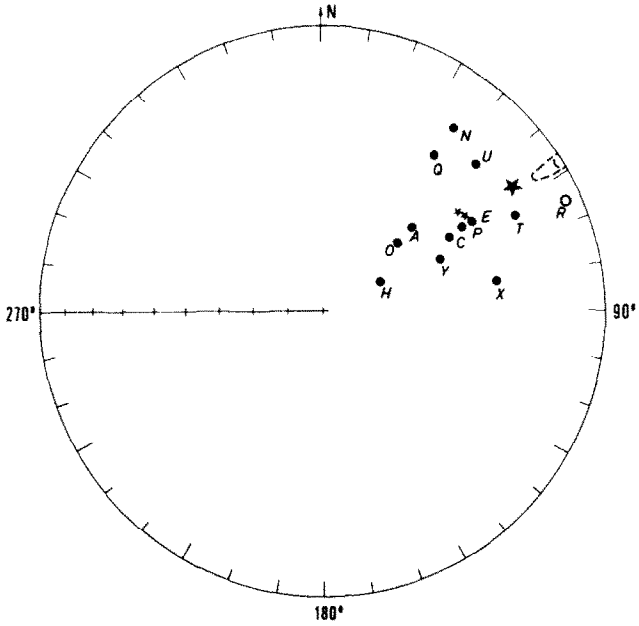


Fig. 4. Directions of remanent magnetization of 13 sites from Sila after AC cleaning. Full (open) symbols = positive (negative) inclinations; small stars = mean directions from 13 sites and from 11 sites (R and H sites excluded); large star = mean direction of 11 sites after correction for Plio—Pleistocene tilt; dashed area = includes the mean direction from 11 sites after a further correction for estimated plunge of the nappes upon emplacement.

sion of the Sila crystalline rocks into two structural units, based upon petrologic and field evidence, is not independently supported by this paleomagnetic data, unless concordant histories of tectonic transport are assumed.

A dip correction can be applied for the post-orogenic Pliocene—Pleistocene block-faulting phases of tectonic activity. Following Ortolani (1976) the entire northeastern Calabrian crustal block, including Sila, was tilted ENE by as much as 20°.

After the tilt correction the mean direction of the Sila crystalline units is $D = 56.5°$ and $I = +20.4°$, with $\alpha_{95} = 9.1°$.

A correction for the overall attitude of the nappes upon emplacement should also be made. A vast majority of authors agree on an easterly to northeasterly plunge for the Sila nappes (eight significant profiles are reported in Ogniben, 1973, pp. 261, 292, 318, 335, 390, 472, 475, 501). Interpretative studies, disregarding these data, show different attitudes: from a westerly plunge of the nappes (Bonardi et al., 1976) to flat-lying bodies (Ogniben, 1969), even to a mushroom-shaped structure apt to conciliate diverging local attitudes (Quitzow, 1935).

By applying the group of corrections suggested by these profiles, inclinations closer to the original lower Permian field are found. In Figs. 4—6 this

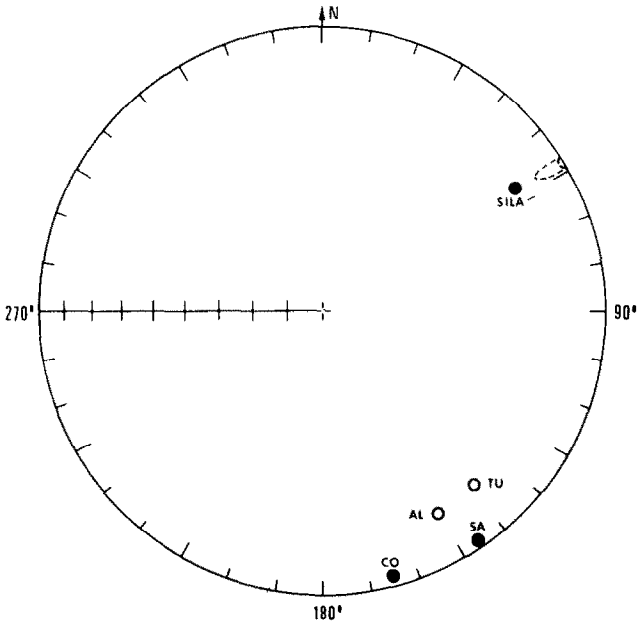


Fig. 5. Directions of remanent magnetization of Lower Permian rocks from Sila, Corsica (*CO*), Sardinia (*SA*), Tuscany (*TU*, Upper Carboniferous—Lower Permian) and Southern Alps (*AL*, Lower—Middle Permian Bolzano porphyries). Dashed area: see Fig. 4.

corrected direction is represented by an area with declinations ranging between 57° and 59° and inclinations between +3.5° and −14°. Low plunges of 20°—35° indirectly confirm the lack of severe disruption, such as would have occurred if relatively thin bodies had been transported, steeply tilted away from sub-horizontal attitudes.

Low positive Early Permian inclinations should be expected for the central northern African margin, and low negative ones for the southern European margin, according to the reconstructions of Zijderveld and Van der Voo (1973) and Van der Voo and French (1974). Since a southern Tethyan domain was advocated as the origin for the chains of the Southern and Western Alps (e.g. Staub, 1951), a comparison of Permian paleolatitudes from the Sila with those for the Southern Alps and for other Mediterranean domains, rather than with continents, is relevant to a discussion of the tectonic history of the Calabrian—Peloritan chain.

Permian inclinations from Sila after corrections are very close to those from Corsica and Sardinia and less close (10°—20° away) to those from the Southern Alps and from Tuscany. This fact could be viewed as further evidence for a non-Apenninic provenance of Sila. Nevertheless, such speculation should be controlled by precise corrections and by more than one-site data from Tuscany.

Comparison of the Permian declinations from Sila with the present-field

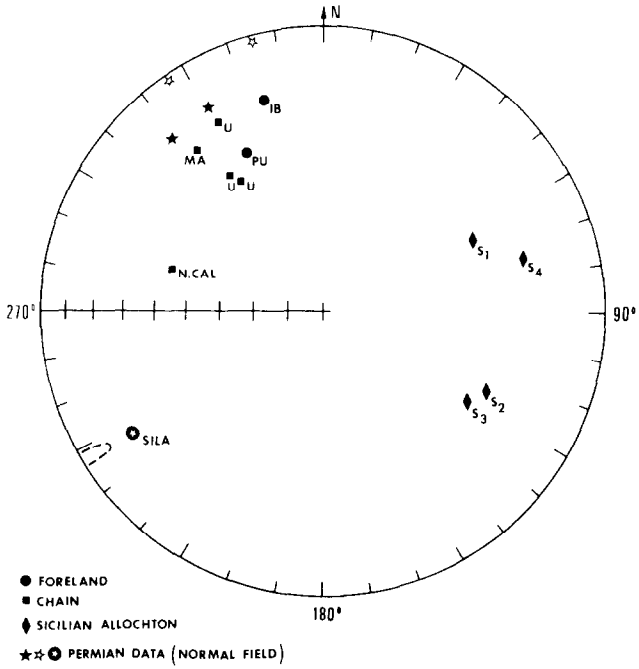


Fig. 6. Directions of magnetization from Sila compared with Lower Permian data of Fig. 5 (stars) and with Upper Cretaceous—Early Tertiary data from the forelands (*IB* = Iblei Mtn, Sicily; *PU* = Apulia), from the Apenninic chain (*U* = Umbria; *MA* = Matese Mtn., Campania; *N. CAL* = Northern Calabria) and from the allochthon of Sicily (S_1, S_2, S_3 = Mt. Bonifato and Imerese basin: S_4 = Custonaci). All directions plotted for a normal field polarity. Dashed area: see Fig. 4.

and paleo-field declinations from surrounding lands is subject to still unsolved geological factors: (1) it has not yet been established whether the basement below the exposed terrains of Calabria is autochthonous or allochthonous, i.e. a drifting crustal plate; (2) strongly divergent hypotheses are proposed for the original locations of the Sila upper nappes and of the other lower nappes of Calabria with metamorphic terrains; and (3) the possible existence of a normal field at the time of cooling of the Sila intrusives cannot be ruled out completely, since a few short normal events occurred during the Kiaman period (Nairn and Peterson, 1973). The probability is low since paleomagnetic data from Sila were gathered from rocks from at least five different intrusive events, in addition to the associated xenoliths and metamorphites. The likelihood that the time-range of all five units was restricted to a normal event of short duration is small. Therefore, the assumption of a reversed field is maintained in the following discussion.

With the assumption of a reversed Permian field, the sense of rotation of the units is counter-clockwise with respect to other Mediterranean lands.

Counter-clockwise rotations of nappes carry the geological significance of

TABLE II

Comparison of the data from Sila with Upper Cretaceous and Early Tertiary data from peninsular Italy and Sicily

Area	Age	D	I	Polarity	Reference
(A) Sicilian allochthon					
Imerese basin	Ku-Pal	119	+37 *	N	Catalano et al. (1976)
Mt. Bonifato	Ku-Pal	63	+41	N	Catalano et al. (1976)
Custonaci	Ku	75	+28	N	Schult (1976)
(B) Foreland					
Iblei (Sicily)	Ku	165	−24 *	R	Schult (1973), Barberi et al. (1974), Gregor et al. (1975)
Apulia	Ku	337	+36	N	Catalano et al. (1976), Channell (1977)
(C) Apenninic chain					
Umbria	Ku	323	+42 *	M	Lowrie and Alvarez (1977b)
Umbria	Pal	326	+42 * (A)	M	Roggenthen and Napoleone (1977)
Umbria	Pal	332	+25 * (B)	M	Roggenthen and Napoleone (1977)
Umbria	Pal-Eo	150	−50 *	R	Vandenberg et al. (1978)
Matese Mts	Ku	322	+29	M	Channell and Tarling (1975)
North Calabria	Ku	287	+44	N	Manzoni (1975b)
(D) Calabria allochthon					
Sila	P_1	57	−7 **	R	this paper

* Recalculated weighted average. The A data by Roggenthen and Napoleone is after thermal cleaning, B data after 150 Oe AC cleaning. Only a representative part of the abundant Umbrian data has been entered in this table; areas with scanty oriented samples have not been included.

** After correction for assumed plunge of the Sila nappes upon emplacement, and correction for Plio-Pleistocene tilt.

TABLE III

Lower Permian paleomagnetic data and pole positions for Central Mediterranean lands, Europe and Africa

Area	Age	*D*	*I*	Polarity	Pole	Reference
Sila *	P_1	56	+20	R	120E 32N	this paper
Sila **	P_1	57	—7	R	132E 22N	this paper
Tuscany	C_u—P_1	338	+21	N	252E 42N	Vandenberg and Wonders (1976)
Dolomites	P_1—P_m	150	—19.5	R	236E 45N	Zijderveld et al. (1970a)
Sardinia ***	P	146	+2	R	238E 37N	Westphal et al. (1976); Zijderveld et al. (1970a)
Corsica ***	P_1	165	+4	R	209E 44N	Westphal et al. (1976); Nairn and Westphal (1968)
Europe	P_1	—	—		161E 44N	McElhinny (1973)
Africa	P_1	—	—		242E 32N	Martin et al. (1978)

* Not corrected for the plunge of the nappes upon emplacement; corrected for recent tilt.

** Corrected for estimated plunge of the nappes upon emplacement and for recent tilt.

*** Recalculated weighted average.

changes of directions during tectonic transport. (The term "rotation" has been most often used to imply motion of large or small crustal plates.) The amount of rotation of Sila calculated from declination differences is about 70° for Corsica, 91° for Sardinia, 94° for the Southern Alps, and 98° for Tuscany. It is certain that rotation was caused by much younger motions; in fact a difference of 105°—110° in declination between Sila and the Sicilian and Apulian forelands postdates the Upper Cretaceous; one of 95° with respect to the Umbrian Apennines postdates the Paleocene and one of 50° with respect to the Apennines of northern Calabria, just across the Sangineto line, postdates the uppermost Cretaceous—Paleogene (see Table II).

The paleomagnetic pole positions for Permian rocks from Sila and from Africa indicate a large relative rotation of 120° (Martin et al., 1978). Lower Permian pole positions relevant to the discussion are given in Table III.

HYPOTHESES OF TECTONIC TRANSPORT

In view of the complex history of the central Mediterranean it is certainly naïve to explain the declination difference between the Sila units and those in other areas by a single phase of displacement.

Most Mediterranean paleomagnetic data fit better with the African pole positions than with those from stable Europe. This leads to the recognition of a general pattern of counter-clockwise motions of the Mediterranean lands with respect to stable Europe. An explanation of such relative rotations lies in a general sinistral component of shear between Africa and Europe, accompanying the compressional phases associated with the collision between the two continents. However, it has been remarked that the rotational movement of Africa was mostly completed by the time main Alpine phases began.

Vandenberg and Wonders (1976) show that the Italian peninsula approximately followed the latitudinal and rotational displacements of Africa between Permian and late Mesozoic times, while various units of the Apennines and of the Southern Alps became uncoupled from each other between the Upper Cretaceous and the Middle—Upper Miocene during the Alpine orogeny. According to Channell and Hovrath (1976) most orogenic activity can be related to paleomagnetically recognized displacements of the African promontory, made by the Iblean platform of Sicily, by Apulia and by the platform of western Greece. Reconstructions of this type cannot yet be attempted for the Calabrian—Peloritan arc since they require data from multiple time-intervals.

The declinational displacement of Sila is compatible with sinistral shear active during the nappes transport with apparent counter-clockwise changes of direction. It is also compatible with the Apenninic transport (the rotation of the Italian peninsula of some authors, e.g., Lowrie and Alvarez, 1975), but Sila data strongly support the occurrence of transport phases with directions different from the vergence of the Apenninic chain; i.e., the Apenninic

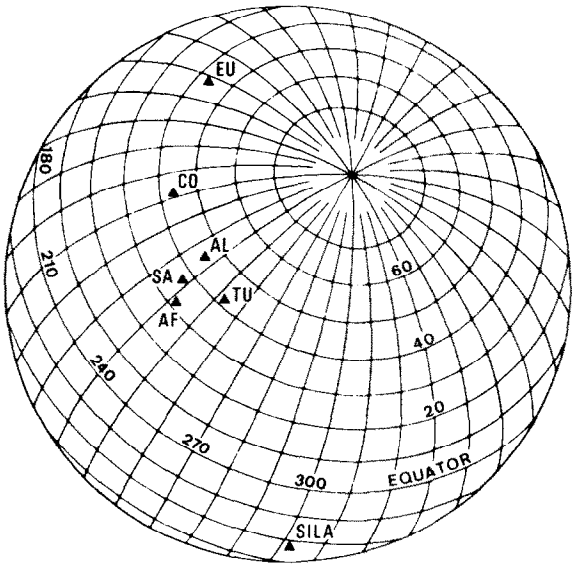


Fig. 7. Lower Permian pole positions calculated from paleomagnetic data from Sila corrected for Plio—Pleistocene tilt, compared with pole-positions from Corsica (*CO*), Sardinia (*SA*), Tuscany (*TU*), the Southern Alps (*AL*), Europe (*EU*; after McElhinny, 1973) and Africa (*AF*, after Martin et al., 1978). South pole position for Sila.

rotation alone cannot account for the total displacement of the Sila nappes and at least one additional phase of displacement must be hypothesized.

The really new implication of the declinational displacement is that directions of tectonic transport must have changed during motion.

Consequently, the pole position calculated for Sila is far from the synchronous African pole positions and is also away from the paleopoles from stable Europe (Fig. 7).

Of the several solutions permitted by paleomagnetic results, two paths of transport are possible considering the structural constraints imposed by Mediterranean geology:

(1) If the last rotational motion, which terminated with the emplacement on the present-day Calabria, was Apenninic, counter-clockwise, from the west, and passively carrying Hercynian nappes with it, this displacement can then be visualized as an arc of motion extending from the west toward the east and subsequently toward the northeast. This motion can be subtracted from the total displacement.

The remaining displacement (older and probably Alpine) requires a different direction of transport along another counter-clockwise arc ending with transport toward the east and starting from the northwest.

This suggested origin strongly recalls the fragmented Alpine chain originally located east of Sardinia and Corsica. The structures with present west-

erly vergence on the coastal chain of Calabria (e.g. Alvarez, 1978), when restored to the orientation preceding the Apenninic transport, acquire roughly northwesterly vergence and, before the pre-Apenninic rotational transport, a roughly northerly vergence. This would be compatible with the frequently assumed northward push of Africa.

(2) Theories advocating an African front require that the final emplacement occurred from the south. The arc of motion can be traced back to an initial phase of transport directed from the west toward the east. This origin lies in the direction of the Hercynian units of the Kabilyes. In this case some east—west parallel transport, probably subsequent to the emplacement of the lower nappes, may have occurred.

Parallel tectonic transport (movement distant from the pole of rotation) cannot be recognized by paleomagnetic methods unless it involves latitudinal displacements larger than the experimental error. It will be undetected by the paleomagnetic method and its amount must be established by geologic and geometric evidence. Changes in the direction of tectonic transport, and movements close to the pole of rotation, on the contrary, can be reconstructed with some confidence.

In both hypotheses the Calabride nappes have not overridden the Iblean platform of Sicily, a stable part of the African promontory. An Ionian provenance is not compatible with the counter-clockwise changes of the direction of tectonic transport indicated by the paleomagnetic data, nor with the sense of displacement of the major fault-zones of the Ionian Sea.

If the assumption of prevailing sinistral shear between Africa and Europe during the orogenic phases is maintained, any eastward motion in the southern Tyrrhenian area would only be compatible with velocities greater than those of the eastward moving African plate, relative to Europe. Locally, the southern border of the rapidly transported tectonic units would give rise to consequent dextral shear lines in superficial tectonics. It could then be hypothesized that the motion of such a border might leave in its wake a swarm of minor superficial clockwise (dextral) swirls: in fact, the allochthonous units of northern Sicily are the only units known to have been rotated clockwise, as indicated by paleomagnetic data (Schult 1976; Catalano et al., 1976; Fig. 8).

Motions along dextral shear lines seem well established elsewhere in the Mediterranean area and often are deemed to be incompatible with paleomagnetic evidence for counter-clockwise motions. In my opinion, the two lines of evidence are not necessarily incompatible, especially when directional changes of tectonic transport are proposed. Whenever the speed of tectonic units uncoupled from the continents is greater than that of the adjoining continent moving in the same direction, a local shear opposite to the overall intercontinental shear trend is generated.

The rotated Italian peninsula seems to be bounded by such dextral lines. The Insubrian line in the Alps is characterized by dextral motion since the Oligocene (Laubscher, 1971), and the "Europe—Africa boundary" by

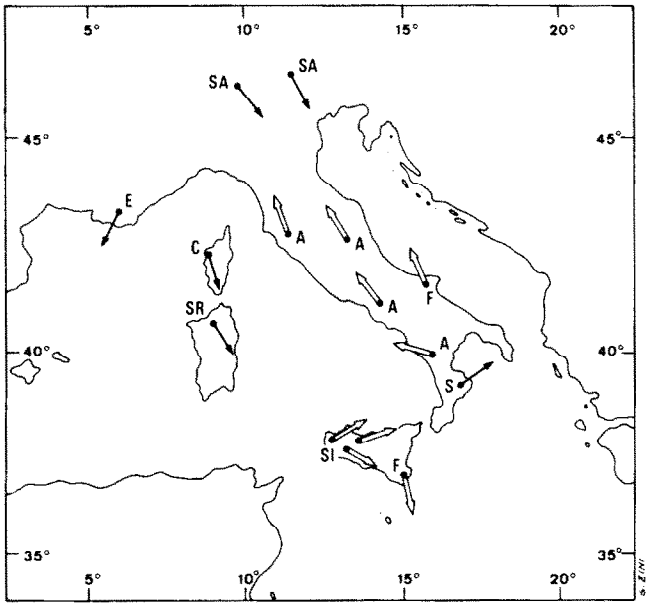


Fig. 8. Paleomagnetic declinations from Sila and the central Mediterranean area. Solid arrows = Permian; Double arrows = Cretaceous—Paleocene. Only declinations from Sila (*S*) and the Sicilian allochthon (*SI*), besides Esterel from "stable" Europe (*E*) are widely divergent from the northwest or southeast trends of the Apennines (*A*), Apenninic foreland (*F*), Southern Alps (*SA*), Corsica (*C*) and Sardinia (*SR*), between Permian and lower Paleogene times.

Barberi et al. (1974) in northern Sicily was considered to be characterized by dextral motion active today. (However, recent research offers evidence for sinistral E—W motions in northern Sicily; Schick, 1977).

As long as the effects of dextral motions are tested in superficial allochthonous covers, such as those of northern Sicily, any discussion on the depth of these shear-zones remains speculative and cannot be taken as evidence for geotectonic hypotheses such as the location of a continental border or the migration of a crustal plate. Thus the paleomagnetic study of the Sila nappes, being part of an allochthonous disrupted basement, is not relevant to the nature of the unexposed crust.

A decisive step toward the reconstruction of the downward extent of non-Apenninic structures in the Calabrian—Peloritan arc might be made by comparing Sila data with data from the lower and intermediate Mesozoic nappes of the coastal chain of Calabria.

The hypothesis of a final westerly or Apenninic provenance of the Sila nappes also requires an active sinistral boundary line allowing for the differences with respect to the data from the Apennines (Manzoni, 1975a and this paper). The Sangineto line can perform such a function: to some French

authors the line even acquires the geotectonic significance of a crustal hiatus (e.g., Caire et al., 1961).

With respect to the Sangineto line, the southern Taormina line can be viewed as the consequent dextral line allowing for the motions of the Calabrian—Peloritan allochthonous units against a passive African promontory. This line would not retain the character of the Africa—Europe boundary, becoming merely the shear-zone between a domain of widely displaced units and a more stable southern Mediterranean foreland, or the African promontory, separated from Europe by intervening minor, independent, once-African domains and by mountain belts.

Finally, in the unlikely event that future study should confirm normal field polarity for the age of formation of the Sila intrusives and therefore clockwise motions of the nappes, the Sangineto line would still remain a main line between major tectonic structures separating a Calabrian—northern Sicilian domain from the Apennines with clockwise motions opposite to those of all other surrounding lands. Thus the history of tectonic transport of the allochthons from Calabria and Sicily would display a unique and constrasting role in respect to the Apenninic foreland and to the Apenninic chain north of the Sangineto line.

Paleomagnetic data carrying the implication of changes of directions during tectonic transport should be taken into account by future reconstructions of the tectonic phases of the Calabrian—Peloritan domain. In fact all directions of stresses active before the Apenninic emplacement of the nappes and all the pre-Middle Miocenic phases of compression and disjunction, as well as the deformational phases affecting older structures, should no longer be referred to the present-day geographic grid, but to paleocoordinates calculated from paleomagnetic data.

The structural trends of Calabria, corrected according to the rotations indicated by paleomagnetic studies, will fall into simpler models and clearer relations between Africa and Europe.

ACKNOWLEDGEMENTS

It is a pleasant duty to acknowledge the help of Dr. A.E.M. Nairn of the University of South Carolina for discussions and for critically reading the manuscript. I also acknowledge the help of Professor S. Lorenzoni and of the team of petrologists of the University of Bari for useful discussions and indications on locations of suitable outcrops.

REFERENCES

Alvarez, W., 1976. A former continuation of the Alps. Geol. Soc. Am. Bull., 87: 891—896.

Alvarez, W., 1978. Calabria as a fragment of the Alpine chain: evidence of nappe transport direction from distorted folds at Monte Reventino (Southern Italy). In press.

Alvarez, W., Cocozza, T. and Wezel, F.C., 1974. Fragmentation of the Alpine orogenic belt by microplate dispersal. Nature, 248: 309—314.

Amodio-Morelli, L., Bonardi, G., Colonna, V., Dietrich, D., Giunta, G., Ippolito, F., Liguori, V., Lorenzoni, S., Paglionico, A., Perrone, V., Piccarretta, G., Russo, M., Scandone, P., Zanettin-Lorenzoni, E. and Zuppetta, A., 1976. L'arco Calabro-peloritano nell'Orogene Appenninico—Maghrebide. Soc. Geol. Ital., 68th Congr.: pp. 1—46.

Barberi, F., Civetta, L., Gasparini, P., Innocenti, F. and Scandone, R., 1974. Evaluation of a section of the Africa—Europe plate boundary: paleomagnetic and volcanological evidence from Sicily. Earth Planet. Sci. Lett., 22: 123—132.

Bemmelen, R.W. Van, 1969. Origin of Western Mediterranean Sea. Verh. Kon. Ned. Geol. Mijnbouwkd. Genoot. Geol. Ser., 51: 548—573.

Boccaletti, M. and Guazzone, G., 1970. La migrazione terziaria dei bacini toscani e la rotazione dell'Appennino Settentrionale in una "zona di torsione" per deriva continentale. Mem. Soc. Geol. Ital., 9: 177—195.

Boccaletti, M. and Manetti, P., 1978. The Tyrrhenian Sea and adjoining Regions. In: A.E.M. Nairn and F.G. Stehli (Editors), Ocean Basins and Margins, 4B. Plenum Press, New York, N.Y., pp. 149—200.

Bonardi, G., Colonna, V., Dietrich, D., Giunta, G., Liguori, V., Lorenzoni, S., Paglionico, A., Perrone, V., Piccarretta, G., Russo, M., Scandone, P., Zanettin-Lorenzoni, E. and Zuppetta, A., 1976. L'arco Calabro-Peloritano. Carta Geologica, 1 : 500.000 map, Soc. Geol. Ital., 68th Congr.

Borsi, S. and Dubois, R., 1968. Données géochronologiques sur l'histoire hercynienne et alpine de la Calabre centrale. C.R. Acad. Sci., 266: 72—75.

Caire, A., Glangeaud, L. and Grandjacquet, C., 1961. Les grands traits structuraux et l'évolution du territoire calabro-sicilien (Italie méridionale). Bull. Soc. Géol. Fr., Ser. 7, 2: 915—938.

Catalano, R., Channell, J.E.T., D'Argenio, B. and Napoleone, G., 1976. Mesozoic paleogeography of the Southern Apennines and Sicily. Problems of paleotectonics and paleomagnetism. Mem. Soc. Geol. Ital., 15: 95—118.

Channell, J.E.T., 1977. Palaeomagnetism of limestones from the Gargano Peninsula (Italy), and the implication of these data. Geophys. J.R. Astron. Soc., 51: 605—616.

Channell, J.E.T. and Horváth, F., 1976. The African—Adriatic Promontory as a palaeogeographical premise for alpine orogeny and plate movements in the Carpatho-Balkan region. Tectonophysics, 35: 71—101.

Channell, J.E.T. and Tarling, D.H., 1975. Palaeomagnetism and the rotation of Italy. Earth Planet. Sci. Lett., 25: 177—188.

D'Argenio, B., Pescatore, T. and Scandone, P., 1973. Schema geologico dell'Appennino meridionale (Campania e Lucania). Atti Accad. Naz. Lincei, Quad., 183: 49—81.

DeFino, M. and LaVolpe, L., 1970. I filoni di Rovale. Rend. Soc. Ital. Mineral. Petrol., 26: 517—546.

De Jong, K.A., Manzoni, M., Stavenga, T., Van Dijk, F., Van der Voo, R. and Zijderveld, J.D.A., 1973. Palaeomagnetic evidence for rotation of Sardinia during the Early Miocene. Nature, 243: 281—283.

Dewey, J.F., Pitman III, W.C., Ryan, W.B.F. and Bonnin, J., 1973. Plate tectonics and the evolution of the Alpine system. Geol. Soc. Am. Bull., 84: 3137—3180.

Dubois, R., 1976. La suture Calabro-Appenninique et l'ouverture Tyrrhénienne Néogène: étude pétrographique et structurale de la Calabre Centrale. Thesis Science, Paris; p. 587.

Dubois, R. and Glangeaud, L., 1965. Grandes structures, microstructures et sens des chevauchements de matériel cristallin à l'extrèmité méridionale du massif de la Sile. C.R. Somm., Séances Soc. Géol. Fr., 7: 229—230.

Gregor, G.B., Nairn, A.E.M. and Negedank, J.F.W., 1975. Paleomagnetic investigations of the Tertiary and Quaternary rocks: IX. The Pliocene of Southeast Sicily and some Cretaceous rocks from Capo Passero. Geol. Rundsch., 64 (3): 948—957.

Haccard, D., Lorenz, C. and Grandjacquet, C., 1972. Essai sur l'évolution tectogénétique de la liaison Alpes—Appennines (de la Ligurie à la Calabre). Mem. Soc. Soc. Geol. Ital., 11: 309—341.

Laubscher, H.P., 1971. The large-scale kinematics of the Western Alps and the Northern Apennines and its palinspastic implications. Am. J. Sci., 271: 193—226.

Lowrie, W. and Alvarez, W., 1975. Paleomagnetic evidence for rotation of the Italian peninsula. J. Geophys. Res., 80: 1579—1592.

Lowrie, W. and Alvarez, W., 1977a. Upper Cretaceous—Paleocene magnetic stratigraphy at Gubbio, Italy. Upper Cretaceous magnetic stratigraphy. Geol. Soc. Am. Bull., 88, (3): 374—377.

Lowrie, W. and Alvarez, W., 1977b. Paleomagnetic studies of the scaglia rossa limestone in Umbria. Mem. Soc. Geol. Ital., 15: 41—50.

Manzoni, M., 1975a. A review of paleomagnetic data from Italy and their interpretations. G. Geol., 39, 2: 513—550.

Manzoni, M., 1975b. Rotation of Calabria: paleomagnetic evidence. Geophys. Res. Lett., 2: 427—429.

Martin, D.L., Nairn, A.E.M., Noltimier, H.C., Petty, M.H. and Schmitt, T.J., 1978. Paleozoic and Mesozoic paleomagnetic results from Morocco. Tectonophysics, 44: 91—114.

Mascle, G. and Grandjacquet, C., 1978. The structure of the Ionian Sea and Calabria-Lucania. In: A.E.M. Nairn and F.G. Stehli (Editors), Ocean Basins and Margins, 4B. Plenum Press, New York, N.Y., pp. 257—329.

Mc Elhinny, M., 1973. Paleomagnetic results from Eurasia. In: D.H. Tarling and S.K. Runcorn (Editors), Implication of Continental Drift to the Earth Sciences. Academic Press, New York, N.Y., 1: 77—85.

Nairn, A.E.M. and Peterson, D.N., 1973. A review of Permian and Triassic paleomagnetic data with respect to paleogeographic conditions at, and the location of, the Permian—Triassic boundary. In: The Permian and Triassic Systems and their Mutual Boundary, pp. 694—713.

Nairn, A.E.M. and Westphal, M., 1968. Possible implications of the palaeomagnetic study of late Palaeozoic igneous rocks of Northwestern Corsica. Palaeogeogr., Paleoclimatol., Paleoecol., 5: 179—204.

Ogniben, L., 1969. Schema introduttivo alla geologia del confine calabro-lucano. Mem. Soc. Geol. Ital., 8 (4): 453—763.

Ogniben, L., 1973. Schema geologico della Calabria in base ai dati odierni. Geol. Rom., 12: 243—585.

Ogniben, L., Parotto, M. and Praturlon, A., 1975. Structural model of Italy (Maps and explanatory notes). Ric. Sci., Quad., 90: 1—502.

Ortolani, F., 1976. Fase tettonica mediopliocenica in Calabria: schema delle principali strutture originate. Boll. Soc. Nat. Napoli, 84: 1—25.

Quitzow, H.W., 1935. Der Deckenbau der Kalabrischen Massiv und seiner Randegebiete. Abh. Ges. Wiss. Math. Phys. Kl., Ser. 3, 13: 63—179.

Roggenthen, W.M. and Napoleone, G., 1977. Upper Cretaceous—Paleocene magnetic stratigraphy at Gubbio, Italy. Upper Maastrichtian—Paleocene Magnetic Stratigraphy. Geol. Soc. Am. Bull., 88: 378—382.

Schick, R., 1977. The focal mechanisms of earthquakes in the Central Mediterranean. Rapp. et P.-V., CIESM, 24 (7): 131—132.

Schult, A., 1973. Palaeomagnetism of Upper Cretaceous volcanic rocks in Sicily. Earth Planet. Sci. Lett., 19: 97—105.

Schult, A., 1976. Palaeomagnetism of Cretaceous and Jurassic volcanic rocks in West Sicily. Earth Planet. Sci. Lett., 31: 454—457.

Staub, R., 1951. Uber die Beziehungen zwischen Alpen und die Gestattung der Alpinen Leittinien Europas. Eclogae Geol. Helv., 44 (1): 29—130.

Vandenberg, J. and Wonders, A.A.H., 1976. Paleomagnetic evidence of large fault displacement around the Po-basin. Tectonophysics, 33: 301—320.

Vandenberg, J., Klootwijk, C.T. and Wonders, A.A.H., 1978. The late Mesozoic and Cenozoic movements of the Italian peninsula: further paleomagnetic data from the Umbrian sequence. Geol. Soc. Am. Bull., 89: 133—150.

Van der Voo, R. and French, R.B., 1974. Apparent polar wandering for the Atlantic bordering continents: late Carboniferous to Eocene. Earth Sci. Rev., 10: 99—119.

Vezzani, L., 1973. L'Appennino Siculo-Calabro-Lucano. Atti Accad. Naz. Lincei. Quad., 183: 1—37.

Westphal, M., Orsini, A. and Vellutini, P., 1976. Le microcontinent Corso-Sarde, sa position initiale: données paléomagnétiques et raccords géologiques. Tectonophysics, 30: 141—157.

Zijderveld, J.D.A. and Van der Voo, R., 1973. Paleomagnetism in the Mediterranean area. In: D.H. Tarling and S.K. Runcorn (Editors), Implication of Continental Drifts to the Earth Sciences, Academic Press, New York, N.Y., 1: 133—161.

Zijderveld, J.D.A., De Jong, K.A. and Van der Voo, R., 1970a. Rotation of Sardinia: Paleomagnetic evidence from Permian rocks. Nature, 226: 933—934.

Zijderveld, J.D.A., Hazeu, G.J.A., Nardin, M. and Van der Voo, R., 1970b. Shear in the Tethys and the Permian palaeomagnetism in the Southern Alps, including new results. Tectonophysics, 10: 639—661.